

Redesigning the World Happiness Report's Country Rankings

Visualization critique and redesign report

Doniyor Erkinov · STATS 401: Data Acquisition and Visualization · Individual project · 28 September 2026

Interactive redesign: donik04.github.io/stats401-labs/individual-project/

1. The original visualization and its context

Figure 2.1 of the *World Happiness Report 2026* (Helliwell et al., 2026, pp. 21–23) ranks 147 countries by their average answer to the Cantril ladder question, from 0 (worst possible life) to 10 (best), using Gallup World Poll data for 2023–2025 (Figure 1). Each country is a horizontal bar labelled with its rank, 95% rank range and score. The bar is split into seven coloured segments: six are the contributions a regression attributes to GDP per capita, social support, healthy life expectancy, freedom, generosity and perceived corruption, measured from “Dystopia”, an imaginary country with the world’s lowest value of every factor; the seventh is “Dystopia (1.16) + residual”.

The figure shows where each country stands, that the ranking is uneven (a crowded top, a long bottom tail, uncertain mid-table ranks), and that six measurable factors explain much of the gap. Its main audience is journalists, policymakers and the public looking up their own country. Readers should be able to look up a country (T1), judge whether a gap is real (T2), see the distribution (T3), compare one factor across countries (T4), read one country’s profile (T5) and find over- and under-performers (T6).

2. Critique

Strengths. The rows are sorted and print the exact rank, rank range and score, so lookup (T1) is precise. And every bar starts at zero on a labelled axis, so total length faithfully encodes the score.

Weaknesses.

- **W1. Stacking defeats comparison.** Only the first segment shares a baseline. The others require comparing unaligned lengths, one of the least accurate perceptual tasks (Cleveland & McGill, 1984), so T4 and T5, the purpose of the decomposition, are effectively impossible.
- **W2. The loudest segment is a constant plus an error term, and it breaks the bar.** The saturated cyan segment is the longest for 38 countries. Because a stack cannot go below zero, a residual below −1.16 makes the bar overshoot: Hong Kong’s reaches 5.87 although its score is 5.57 (Figure 2).

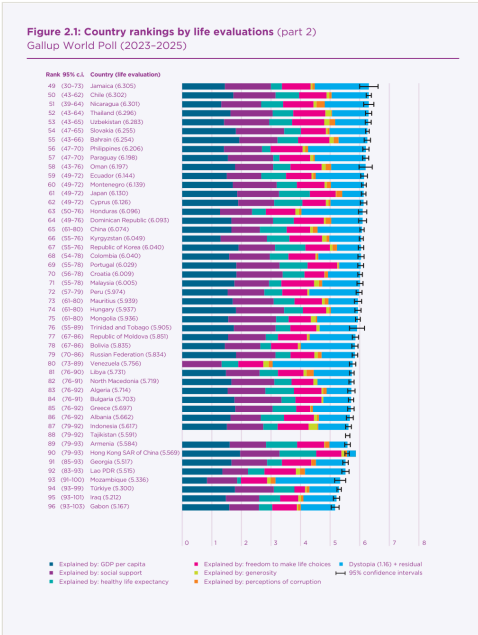


Figure 1. The original Figure 2.1, part 2 of 3 (ranks 49–96), *World Happiness Report 2026*, p. 22. Source: Helliwell et al. (2026).

- W3. The zero point is abstract and fragile.** Dystopia’s GDP comes from Venezuela, listed in the Statistical Appendix at \$89 per person (Yemen, next lowest, is at \$917). Every GDP segment is measured from that value, and Venezuela gets the world’s largest residual.
- W4. Gaps and uncertainty are hidden.** Tajikistan and Palestine get empty rows although each lacks only one factor. Rank ranges are bracketed text, so the report’s own point that mid-table ranges span 20–44 places is invisible, and the list is split across three pages.

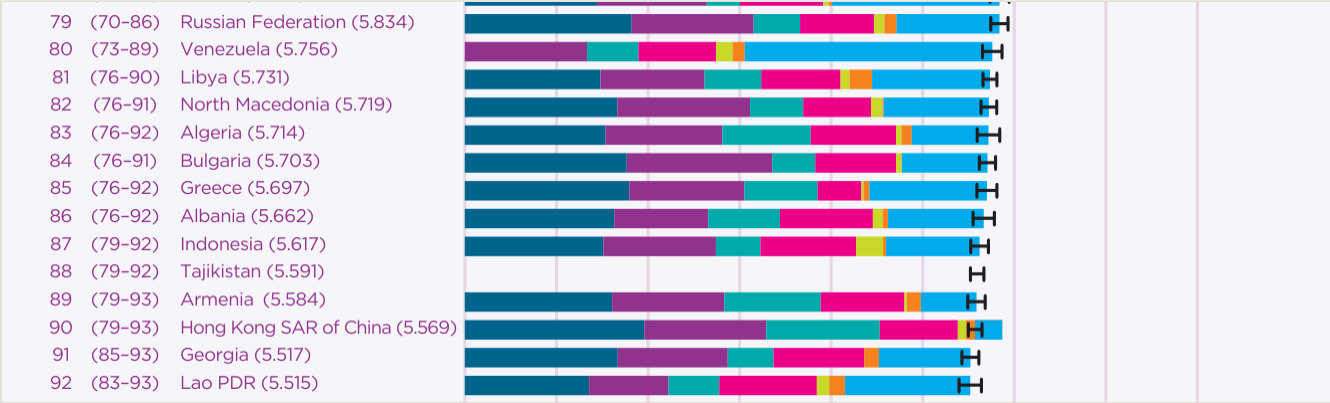


Figure 2. Ranks 79–92 of the original, enlarged. Venezuela has no GDP segment (W3), Tajikistan’s row is empty (W4), and Hong Kong’s bar runs past its own confidence whisker (W2).

3. Redesign rationale

The redesign uses the report’s own data in four linked views; clicking a country in any view selects it (Figure 3).

D1. A waterfall from the average country (W2, W3; T5, T6). A country’s pieces become steps on separate rows, running from the average country’s predicted score (5.65) to the actual score: blue where a factor raises the prediction, red where it lowers it, and the residual as the final step. A waterfall keeps the stack’s one real virtue, parts that add up to the whole, while allowing negative parts. The average country is a benchmark readers understand and, unlike Dystopia, it ignores the \$89 value. Hong Kong’s factors now climb to 7.03 before the unexplained step visibly drops it to 5.57 (Figure 4). A toggle restores the report’s baseline.

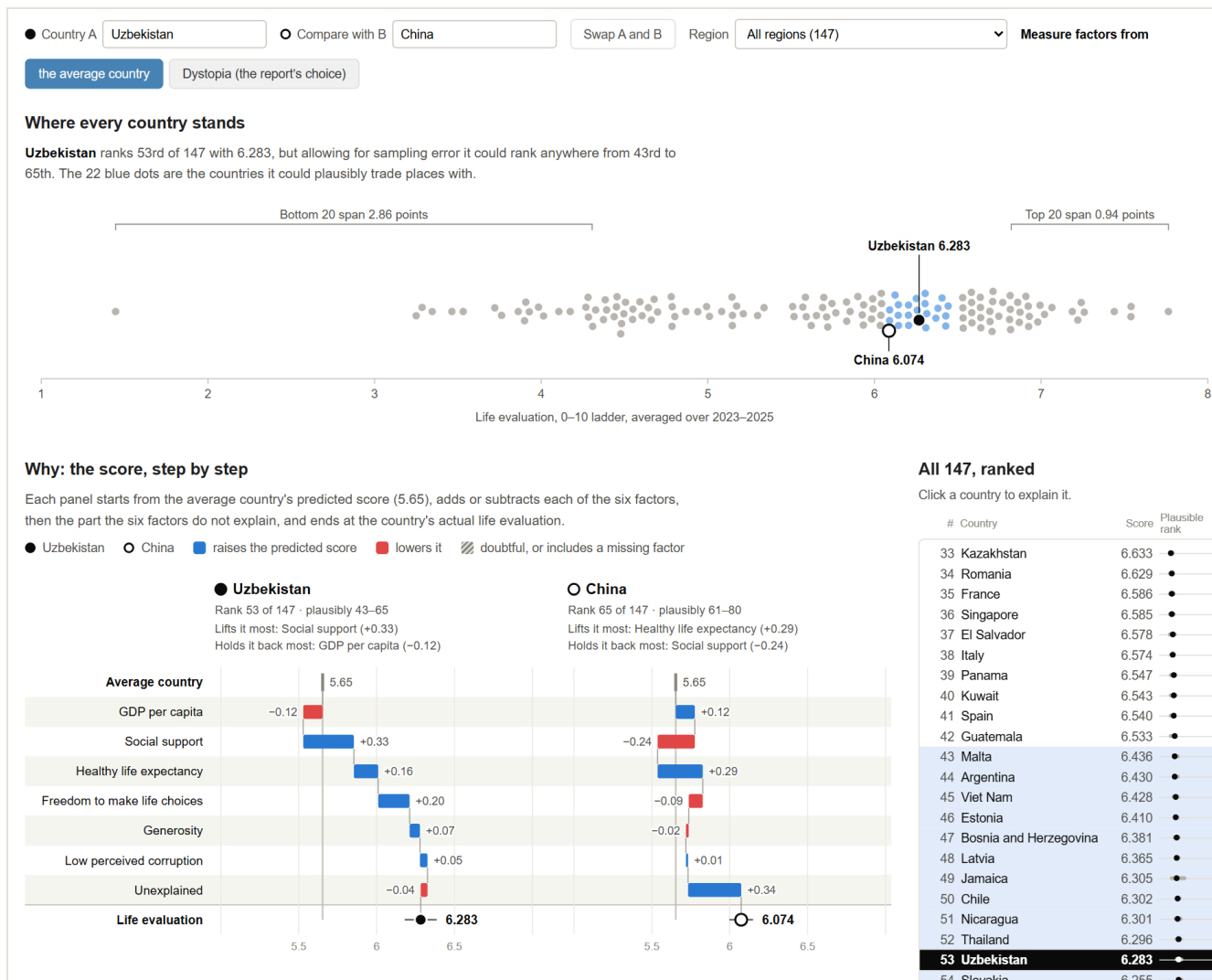


Figure 3. The redesign, with Uzbekistan as country A and China as B: the controls, the beeswarm overview (the countries Uzbekistan could plausibly trade places with are blue), the two waterfalls and the ranked list.

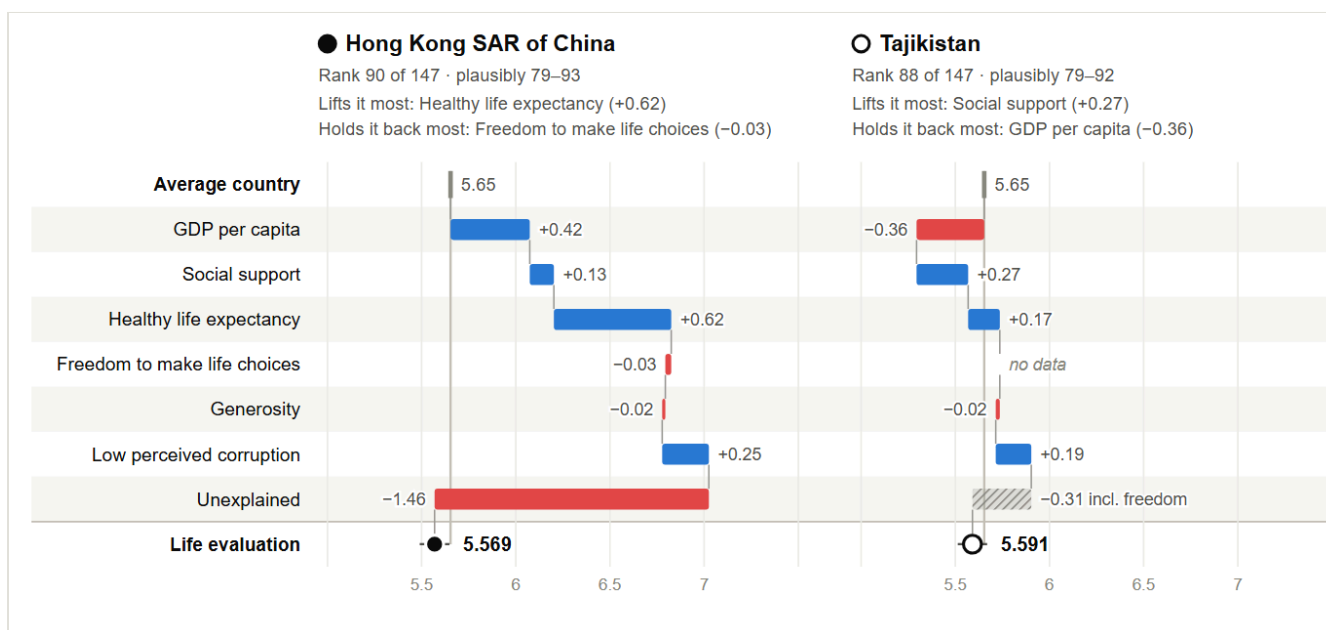


Figure 4. Hong Kong's unexplained drop, which the original drew as an overshoot, and Tajikistan's missing freedom factor, shown as “no data” and absorbed by a hatched final step.

D2. Strip plots on one shared scale (W1; T4, T6). Each factor gets its own axis with all 147 countries, so comparisons use aligned position. The shared scale also shows how much each factor matters: GDP spans 2.2 points, but generosity spans only 0.3 (Figure 5).

D3. Uncertainty as a group, distribution as a shape (W4; T2, T3). A beeswarm puts every score on one axis and measures the spread of the top and bottom 20; the countries inside the selected country's rank range turn blue.

D4. Honest gaps and one navigable list (W1, W4; T1). Missing factors say “no data” and their effect becomes a hatched final step; doubtful values are hatched and footnoted. One ranked list with search and a region filter replaces the three pages, and colour encodes only direction.



Figure 5. Every country on each factor, on one shared scale in ladder points relative to the average country. Uzbekistan is the filled dot and China the ring.

4. Original versus redesign

Tasks the original blocked are now direct. Comparing Uzbekistan (6.28) with China (6.07) shows why one outranks the other: social support adds 0.33 for Uzbekistan and subtracts 0.24 for China. Outliers, rank uncertainty and regional comparison each take one click, and lookup is as easy as before.

The costs: only two countries are explained at a time, where the original shows all 147 at once. Waterfall steps float between panels, so their values are printed. Measuring from the average departs from the report's numbers. The design also needs a screen, and the decomposition stays correlational, not causal.

References

- Bostock, M. (2026). *D3.js* (Version 7) [Computer software]. <https://d3js.org>
- Cleveland, W. S., & McGill, R. (1984). Graphical perception: Theory, experimentation, and application to the development of graphical methods. *Journal of the American Statistical Association*, 79(387), 531–554. <https://doi.org/10.1080/01621459.1984.10478080>

- Helliwell, J. F., Aknin, L. B., Huang, H., Rojas, M., Wang, S., Guerra, V., & Danyluk, A. (2026). *Statistical appendix for "International evidence on happiness and social media," Chapter 2 of World Happiness Report 2026*. https://files.worldhappiness.report/WHR26_Statistical_Appendix.pdf
- Helliwell, J. F., Layard, R., Sachs, J. D., De Neve, J.-E., Aknin, L. B., & Wang, S. (Eds.). (2026). *World happiness report 2026*. Wellbeing Research Centre, University of Oxford. <https://files.worldhappiness.report/WHR26.pdf>
- World Happiness Report. (2021). *Data for Figure 2.1* [Data set; used for the regional grouping]. https://files.worldhappiness.report/WHR21_Data_Figure_2.1.xls
- World Happiness Report. (2026). *Data for Figure 2.1* [Data set]. https://files.worldhappiness.report/WHR26_Data_Figure_2.1.xlsx